

Values concerning item bank building (item linking, -calibration, equating)

	item bank building (evaluation)			result
	testforms	items/form	examinees	items/bank
matrice test	76	12	29.728	456
analogy test	25	24	12.548	254
arithmetic test	32	18	17.008	288

In a further step the adaptive 2 PL-algorithm was evaluated in combination with the three item banks mentioned above. Hence 8000 examinees (draftees) worked on the usual conventional test battery. Additionally, they received one of the three tests which corresponded to one test in the test battery. The additional test was administered as an adaptive one.

The aim of adaptive testing in the testing situation is to gain maximum information with minimum effort. Hence the examinees should be given only those items that meet the person's level of functioning, so that the items will be solved in about fifty percent of the cases. If that holds, irrelevant items can be economized on, and, as one of various merits, it can be expected that adaptive testing, particularly the 2 PL, will need much less items as against conventional tests based on classical test theory. The conventional tests of our test battery consist of 20 items each with moderate reliability ($r_{tt} = 0.74$ to $r_{tt} = 0.88$). Figure 1 is based on the compound result of the three tests. It shows the joint distribution of the adaptive items needed to reach a reliability of $r_{tt} \gg 0.84$ for each person. This reliability is associated with a standard error of measurement of $SEM = 0.39$ for this sample. With the appearance of variance in the answer vector, the SEM is calculated whenever a person answers an administered item within the adaptive testing procedure. Before testing, the SEM value is fixed on a corresponding desired reliability level (in this case $r_{tt} = 0.84$). This cut-off value serves as stop criterion of the adaptive procedure as soon as the SEM falls below.